Multivariate Time-Series Prediction Using Deep Learning

Subject: Appliance Energy Prediction Using the Energy Dataset

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1. Introduction

Electrical energy consumption prediction is an important task in energy management, as it helps improve energy efficiency, reduce electricity costs, and support better resource planning. Accurate forecasting of household appliance energy consumption can help identify usage patterns and provide insights into how environmental and time-related factors influence electricity demand.

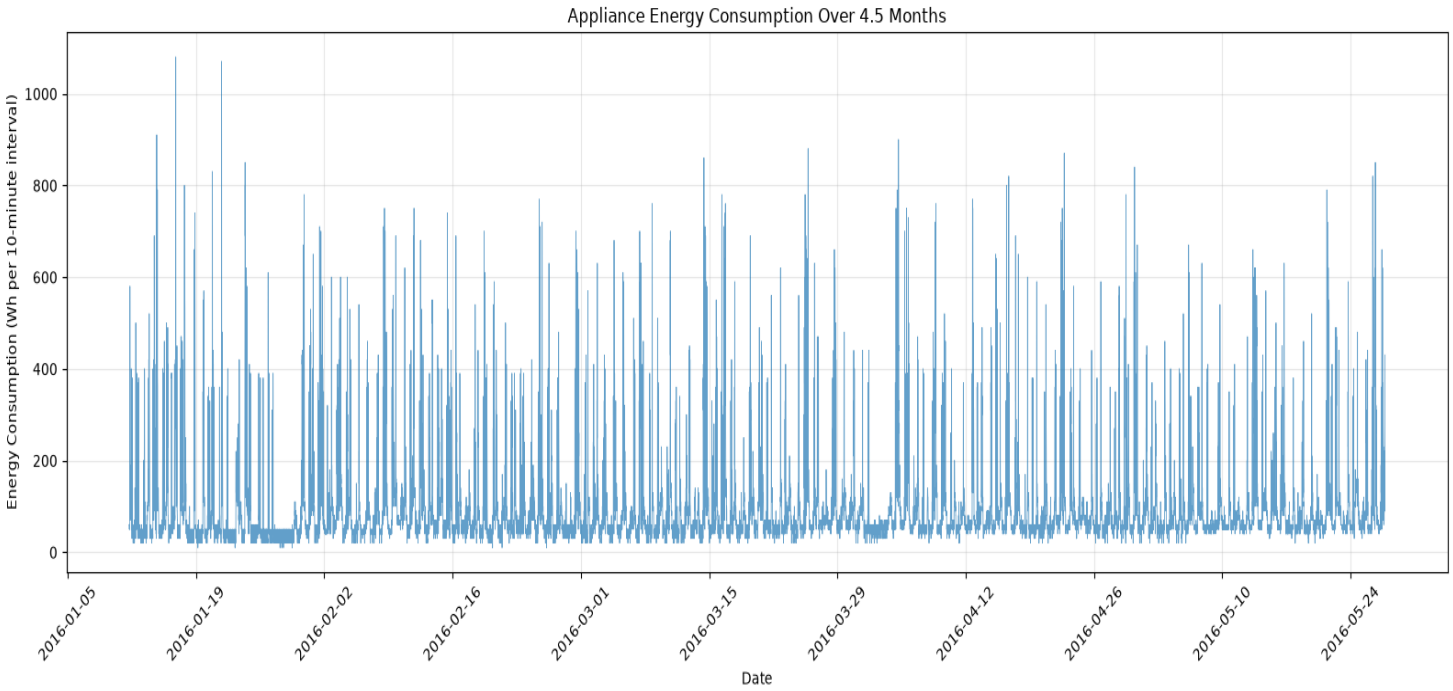
This project focuses on predicting appliance energy consumption using the Appliances Energy Prediction Dataset, which contains multivariate time-series data collected at 10-minute intervals over several months. The dataset includes appliance energy consumption measurements, indoor and outdoor temperature, humidity, lighting, wind speed, atmospheric pressure, and other environmental and time-related variables.

The main objective of this project is to develop and evaluate deep learning models capable of forecasting future appliance energy consumption based on historical observations and associated environmental conditions. Several approaches, including Long Short-Term Memory (LSTM), Convolutional LSTM (CNN-LSTM), and hybrid deep learning architectures, are explored to understand their ability to capture temporal patterns and relationships within the data.

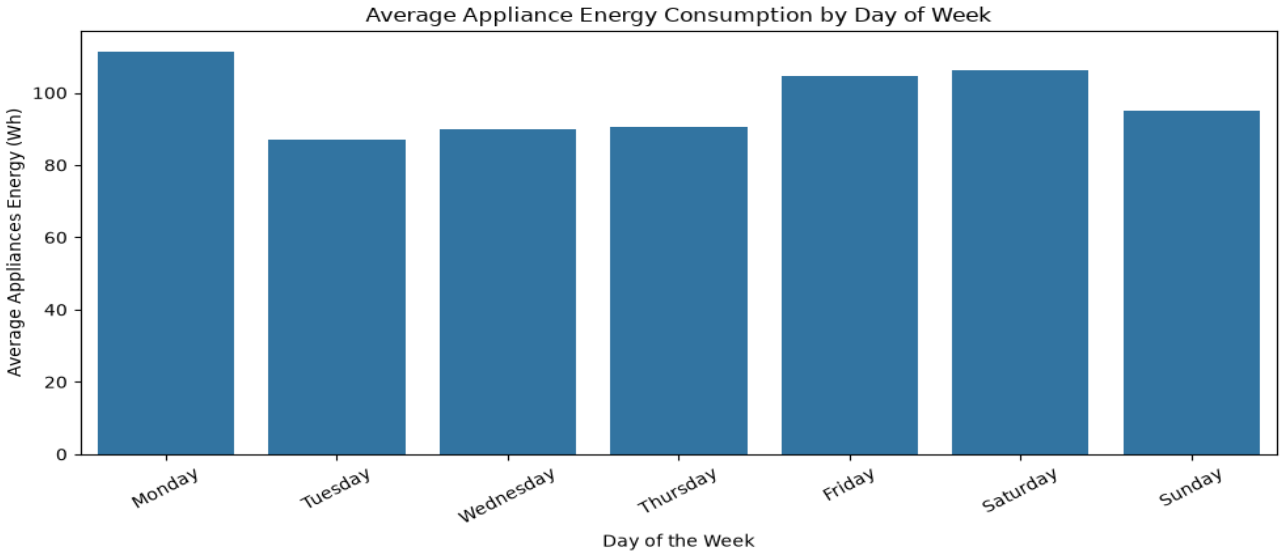
The project involves data preprocessing, exploratory data analysis, feature engineering, model development, training, and performance evaluation. The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Ultimately, this project aims to identify a suitable deep learning approach for appliance energy consumption forecasting and assess how effectively historical energy usage and environmental information can be used to predict future consumption.

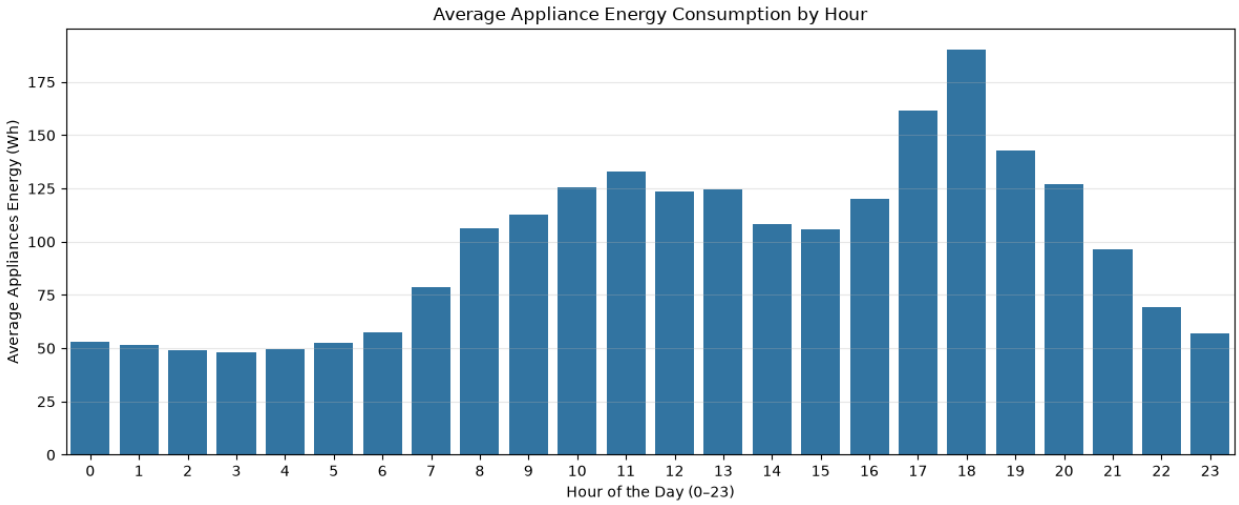
2.Data Insights



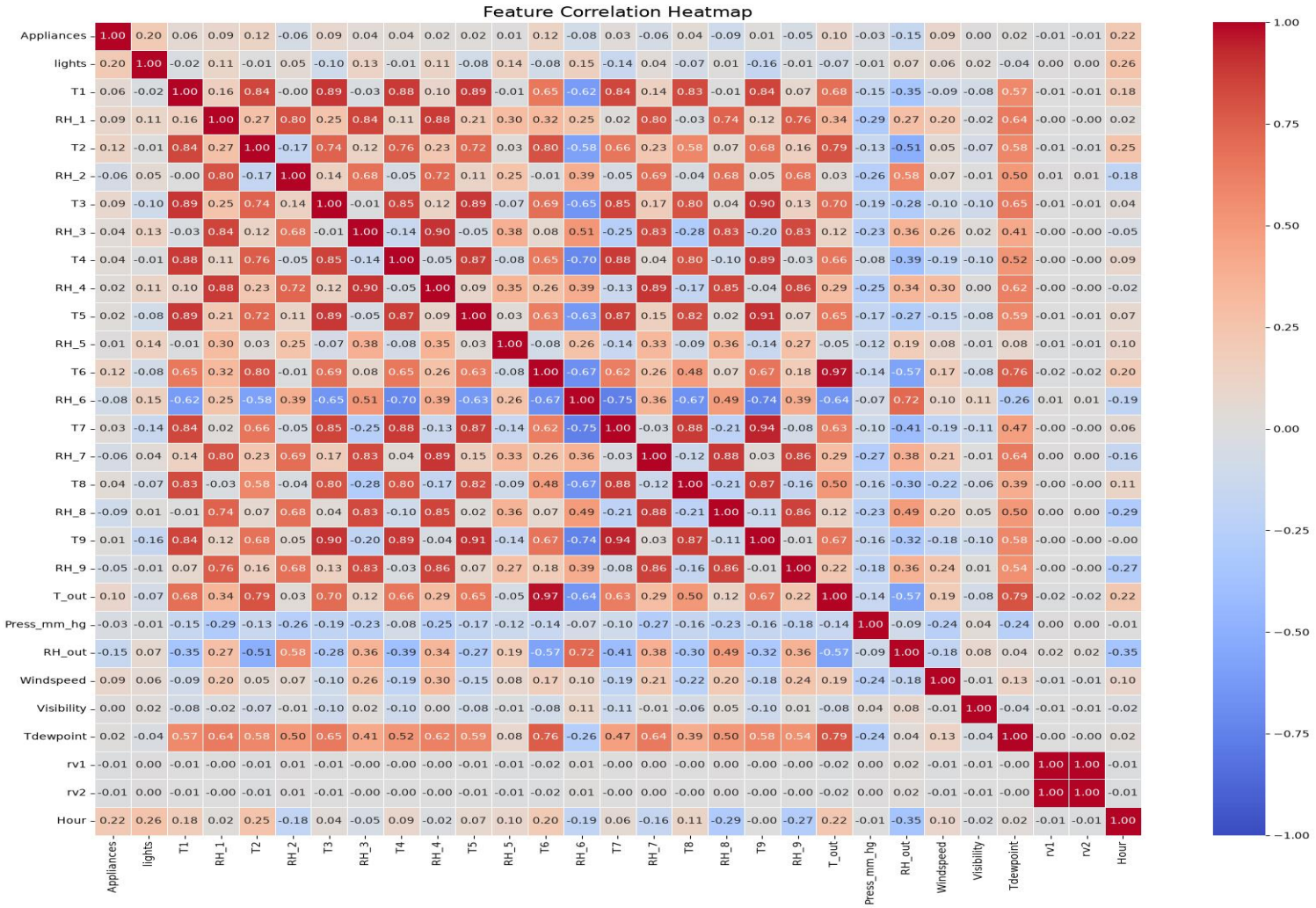
The appliance energy consumption plot reveals several important patterns, particularly sudden peaks and fluctuations in energy usage. These observations suggest that previous consumption values can influence future energy consumption. Therefore, historical features such as lag values, gradients (rate of change), rolling averages, and rolling maximum values may provide useful information for predicting future consumption. In particular, capturing sudden increases or decreases in energy usage could help the models better understand consumption patterns and improve forecasting accuracy



This figure indicates that energy consumption varies across different days, with certain days experiencing relatively higher usage. Therefore, incorporating day-related features, such as the day of the week and weekday/weekend indicators, could help the model capture these recurring patterns and improve prediction accuracy.



The hourly energy consumption pattern also reveals variations throughout the day, with relatively lower consumption during daytime and higher consumption during nighttime. This reflects daily household usage patterns. Therefore, incorporating time-related features, particularly the hour of the day, can help the model capture these recurring patterns and improve forecasting accuracy.



The correlation matrix reveals several interesting relationships between the dataset features. A negative correlation can be observed between temperature and humidity measurements, reflecting the inverse relationship that can occur between these environmental factors. As observed in the earlier analysis, time-related features, particularly the hour of the day, also show meaningful correlations with energy consumption. In contrast, features such as visibility exhibit relatively weak correlations. Additionally, the random variables (rv1 and rv2) show very little correlation with other features, as expected. These observations provide useful insights for understanding feature relationships and identifying potentially valuable inputs for the prediction models.

3.Preprocessing

During data preprocessing, the dataset was checked for missing values, and no missing values were found. The time-related features were also transformed into sine and cosine representations to preserve their cyclical nature and help the models capture recurring time-based patterns. Outlier detection was then performed using the Z-score method. The figure below highlights the observations identified as potential outliers

Feature	Mean	Standard Deviation	Outlier Count	Outlier Percentage (%)
lights	3.802	7.936	647	3.28
RH_5	50.949	9.022	567	2.87
Appliances	97.695	102.525	540	2.74
T2	20.341	2.193	206	1.04
RH_out	79.75	14.901	159	0.81
RH_2	40.42	4.07	131	0.66
RH_1	40.26	3.979	109	0.55
Press_mm_hg	755.523	7.399	107	0.54
Windspeed	4.04	2.451	104	0.53
T6	7.911	6.09	98	0.5
T_out	7.412	5.317	93	0.47
T3	22.268	2.006	54	0.27
Visibility	38.331	11.795	37	0.19
RH_7	35.388	5.114	24	0.12
T5	19.592	1.845	23	0.12
RH_3	39.243	3.255	14	0.07
T1	21.687	1.606	14	0.07
RH_8	42.936	5.224	7	0.04
T4	20.855	2.043	0	0.0
RH_4	39.027	4.341	0	0.0
T8	22.029	1.956	0	0.0
RH_6	54.609	31.15	0	0.0
T7	20.267	2.11	0	0.0
T9	19.486	2.015	0	0.0
RH_9	41.552	4.151	0	0.0
Tdewpoint	3.761	4.195	0	0.0
Hour	11.502	6.922	0	0.0

Outlier Analysis and Handling

Although the Z-score method identified potential outliers in several features, further investigation suggests that most of these values are genuine observations rather than data errors.

- Lights: The Z-score method identifies some lighting values as outliers because the feature contains a large number of zero readings. However, values such as 30 and 40 are realistic, particularly during nighttime. Even higher readings, such as 60 and 70, occur occasionally and are also possible. Therefore, these values are considered genuine and are retained.

- Appliances: Although some high energy consumption readings are flagged as potential outliers, the consumption increases relatively smoothly rather than showing unexpected sudden spikes. Furthermore, these values are practically possible in real-world household energy usage. Hence, they are retained.

- Temperature and Humidity: The flagged values fall within plausible environmental conditions and reflect natural variations. Therefore, there is insufficient evidence to classify them as erroneous observations.

- Atmospheric Pressure and Wind Speed: Potential outliers in these features appear to occur together. Since atmospheric conditions and wind speed are related, these observations may reflect genuine environmental changes rather than measurement errors. Therefore, they are retained.

Overall, although the Z-score method identified several potential outliers, examining their distributions, temporal patterns, and real-world plausibility suggests that they are mostly genuine observations. Removing them could result in the loss of valuable information. Consequently, no outliers were removed, and the original dataset was retained for subsequent preprocessing and model development.

Data Scaling and Normalization

Standardization was selected instead of Min-Max Scaling because the dataset contains some extreme values. Min-Max Scaling could compress most frequently

occurring values into a narrow range, making differences between them less distinguishable. Standardization reduces the influence of these extreme values on the scaling range while preserving the relative variation in the data.

4.Feature Engineering

Based on the exploratory data analysis and correlation matrix, additional features were created to capture temporal patterns and historical energy consumption behaviour. The objective was to provide the deep learning models with more meaningful information about previous consumption patterns rather than relying only on the original dataset features.

The following feature groups were introduced:

1. Time-based features: Features such as day of the week, weekday/weekend status, and hour of the day were included to capture recurring daily and weekly consumption patterns. Sine and cosine transformations of the hour were also introduced to represent the cyclical nature of time.

2. Lag features: Previous appliance consumption values were included at different intervals, such as 10 minutes, 30 minutes, 1 hour, 3 hours, and 24 hours. These features help identify how historical energy consumption influences future usage.

3. Rolling statistical features: Rolling mean, standard deviation, minimum, and maximum values were calculated over 1-hour and 3-hour windows. These features help capture recent consumption trends, fluctuations, and variations in energy usage.

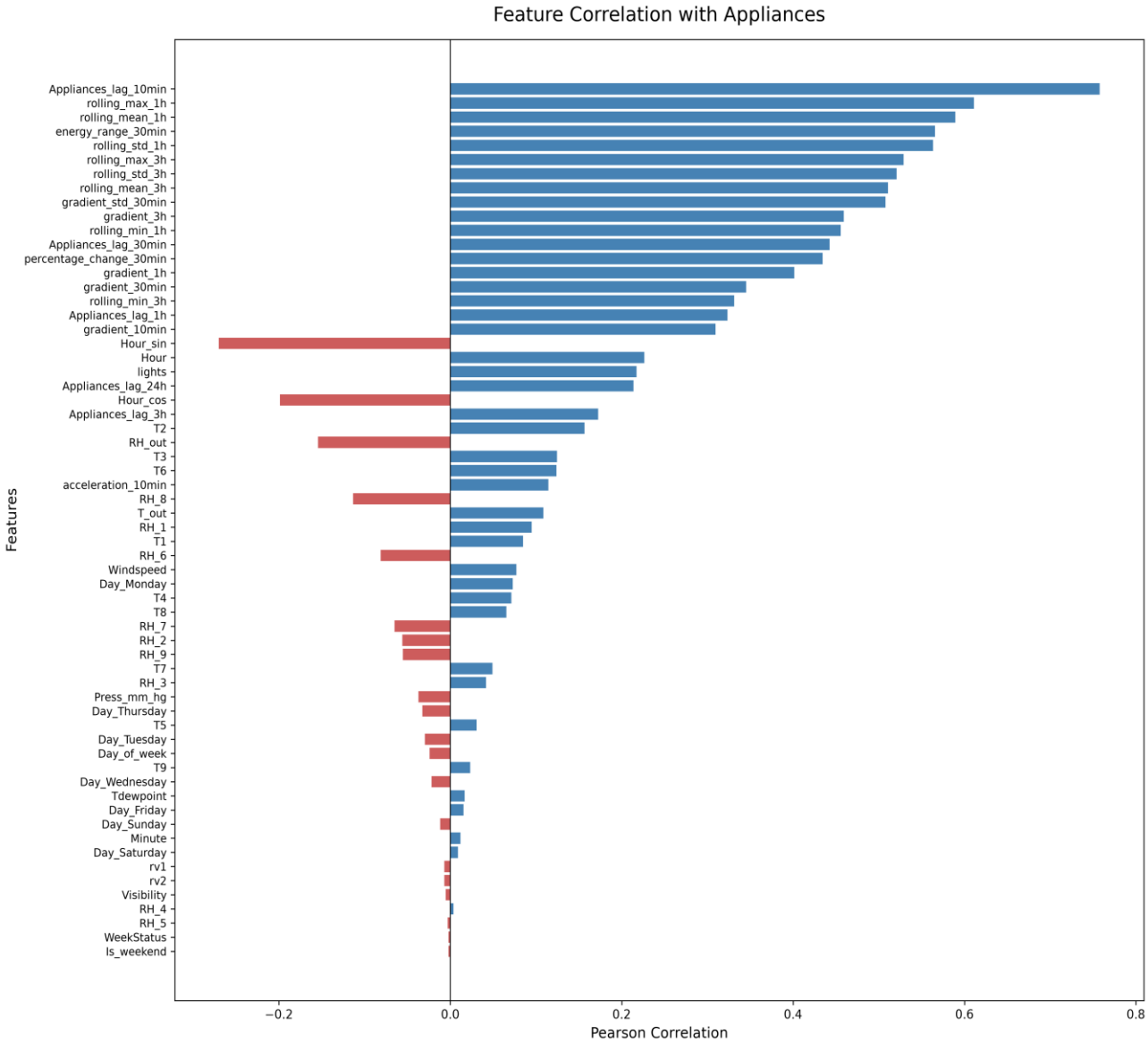
4. Gradient features: Consumption gradients were calculated over 10-minute, 30-minute, 1-hour, and 3-hour intervals to represent the rate of change in energy consumption. These features help identify increasing and decreasing consumption trends.

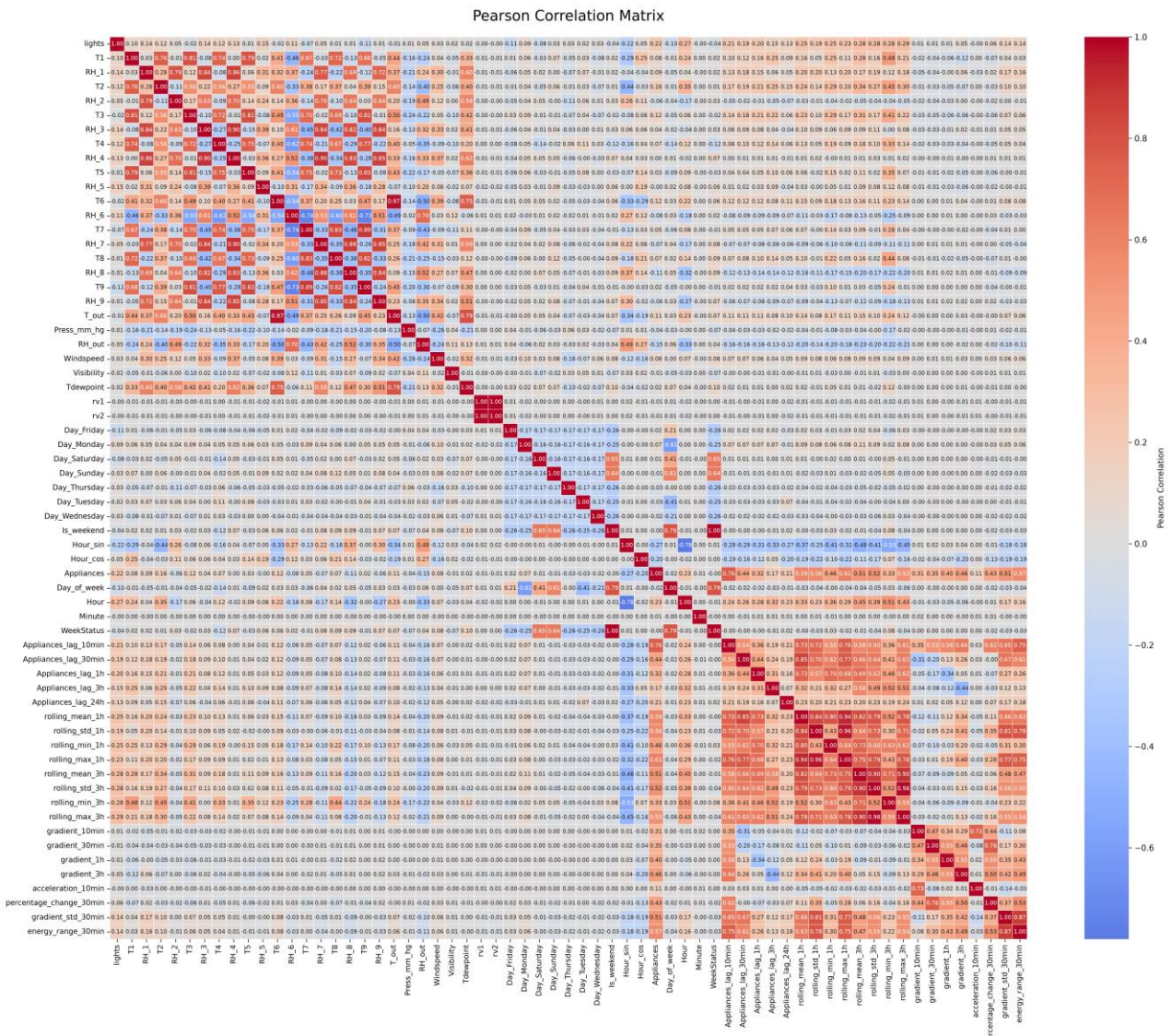
5. Additional statistical features: Acceleration, percentage change, rolling gradient standard deviation, and energy range were introduced to capture sudden changes, fluctuations, and variations in historical consumption.

Feature Selection

Different time intervals were considered when generating lag, rolling statistical, and gradient features, including 10 minutes, 30 minutes, 1 hour, 3 hours, and 24 hours. The correlation matrix was then examined to understand the relationships between the engineered features and the target variable, Appliances.

Based on these observations, potentially informative features were considered for inclusion in the final dataset. This approach helps retain useful historical information while avoiding unnecessary features that provide limited predictive value.





Feature selection for liner Regression and Deep learning models

Following feature engineering, feature selection was performed to identify the most relevant input features while reducing redundancy among highly correlated variables. The correlation matrix was examined to understand both the relationships between the engineered features and the target variable (Appliances) and the correlations between the input features themselves.

Although some features, particularly those calculated over 1-hour and 3-hour windows, showed good correlations with the target, several were also strongly correlated with features calculated over shorter intervals. Since these features could

contain overlapping information about historical energy consumption, some were excluded to reduce redundancy and maintain a more compact input feature set.

he following features were selected as model inputs for training and prediction:

date, lights, T1, RH_1, T2, RH_2, T3, RH_3, T4, T5, T6, RH_6, T7, RH_7, T8, RH_8, T9, RH_9, T_out, Press_mm_hg, RH_out, Windspeed, Tdewpoint, Day_Friday, Day_Monday, Day_Saturday, Day_Sunday, Day_Thursday, Day_Tuesday, Day_Wednesday, Hour_sin, Hour_cos, Appliances, Day_of_week, Hour, Appliances_lag_10min, rolling_mean_1h, rolling_std_1h, rolling_min_1h, and rolling_max_1h.

These selected features were retained for model development, while the remaining features were excluded from the model inputs.

The remaining features were retained as inputs for both the Linear Regression and deep learning models. These included the original environmental and time-related variables, along with the selected engineered features, such as shorter-window lag features, rolling statistics, gradients, acceleration, and cyclical time representations.

The same selected feature set was used for both Linear Regression and deep learning models to maintain consistency when comparing their predictive performance.

Random Forest Feature Selection

For the Random Forest model, feature selection was performed using the feature importance scores obtained from an initial Random Forest Regressor trained on all available numerical input features. The features were ranked according to their importance, and the top 15 features were selected as inputs for the final model. The selected features were **Appliances_lag_10min, acceleration_10min, gradient_10min, Hour_sin, rolling_mean_3h, Hour_cos, rolling_std_3h, Appliances_lag_30min, rolling_std_1h, Press_mm_hg, rolling_mean_1h, Appliances_lag_24h, T4, gradient_1h, and RH_5**. This approach helped identify the most influential features and reduce the input feature set from all available numerical features to 15.

5.Model Design

Long Short term Memory (LSTM)

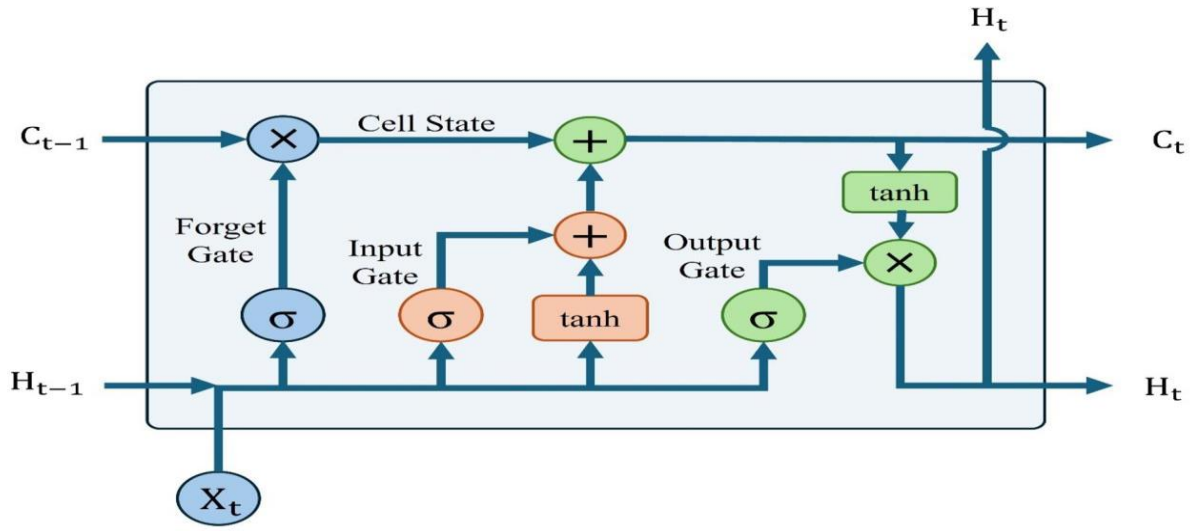
Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network (RNN) designed to learn long-term dependencies in sequential data. Unlike traditional neural networks, LSTMs maintain information from previous time steps through an internal memory mechanism called the cell state. This makes them particularly suitable for time-series forecasting, where future values may depend on historical patterns.

LSTM Architecture

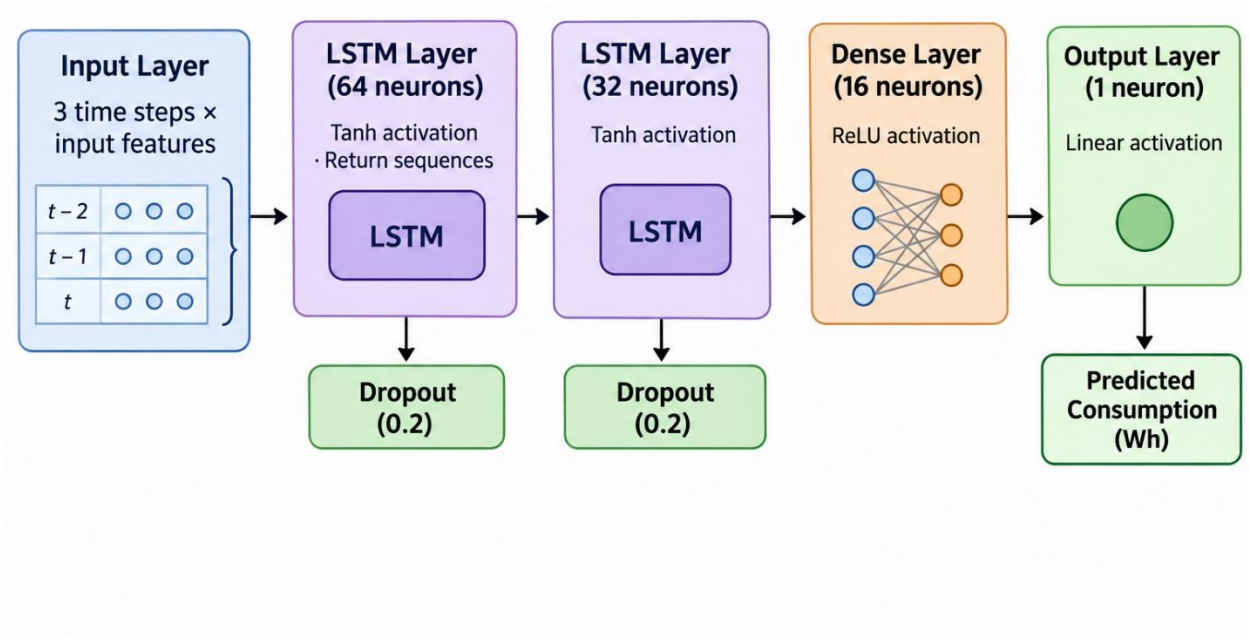
An LSTM cell consists of three main gates that control the flow of information:

1. Forget Gate: Determines which information from the previous cell state should be retained or dis carded.
2. Input Gate: Determines which new information should be added to the cell state.
3. Output Gate: Determines which information from the cell state should be used to generate the current hidden state.

The cell state maintains long-term information, while the hidden state carries the relevant information to the next time step. These mechanisms allow LSTMs to learn complex temporal patterns and reduce the vanishing gradient problem associated with traditional RNNs. [1]



Initial Hyperparameter Values



Tanh: Used in both LSTM layers to regulate information flow through the recurrent units, producing values between −1 and 1.

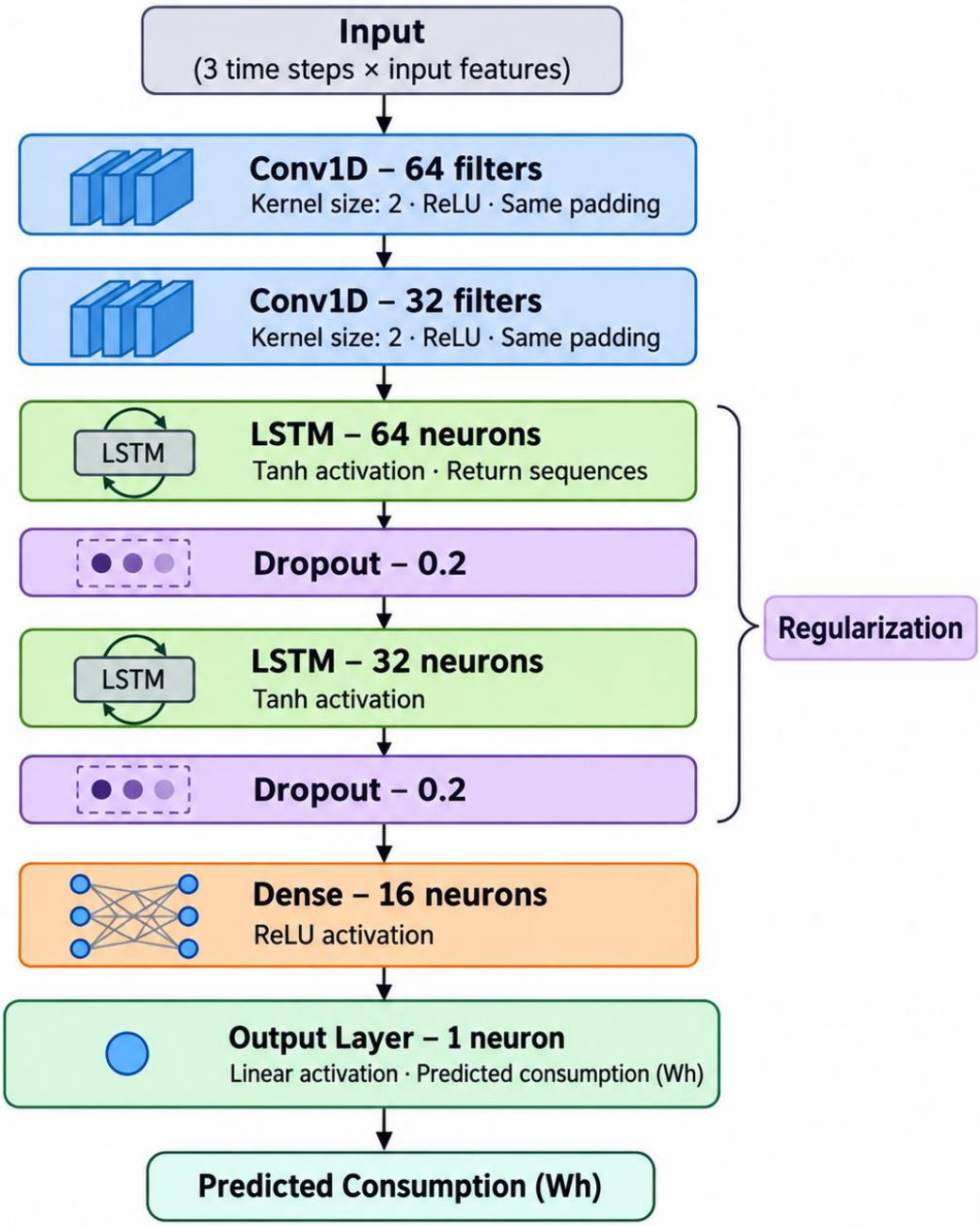
ReLU: Used in the Dense layer to introduce non-linearity and learn complex relationships.

Linear: Used in the output layer to predict continuous energy consumption values.

Adam Optimizer: Selected for its adaptive learning rates and momentum-based updates, enabling efficient model weight optimization.

CNN-LSTM Model Architecture

The CNN-LSTM model combines Convolutional Neural Network (CNN) layers and Long Short-Term Memory (LSTM) layers to extract local patterns and learn temporal dependencies from historical energy consumption data [2].



Layer justification

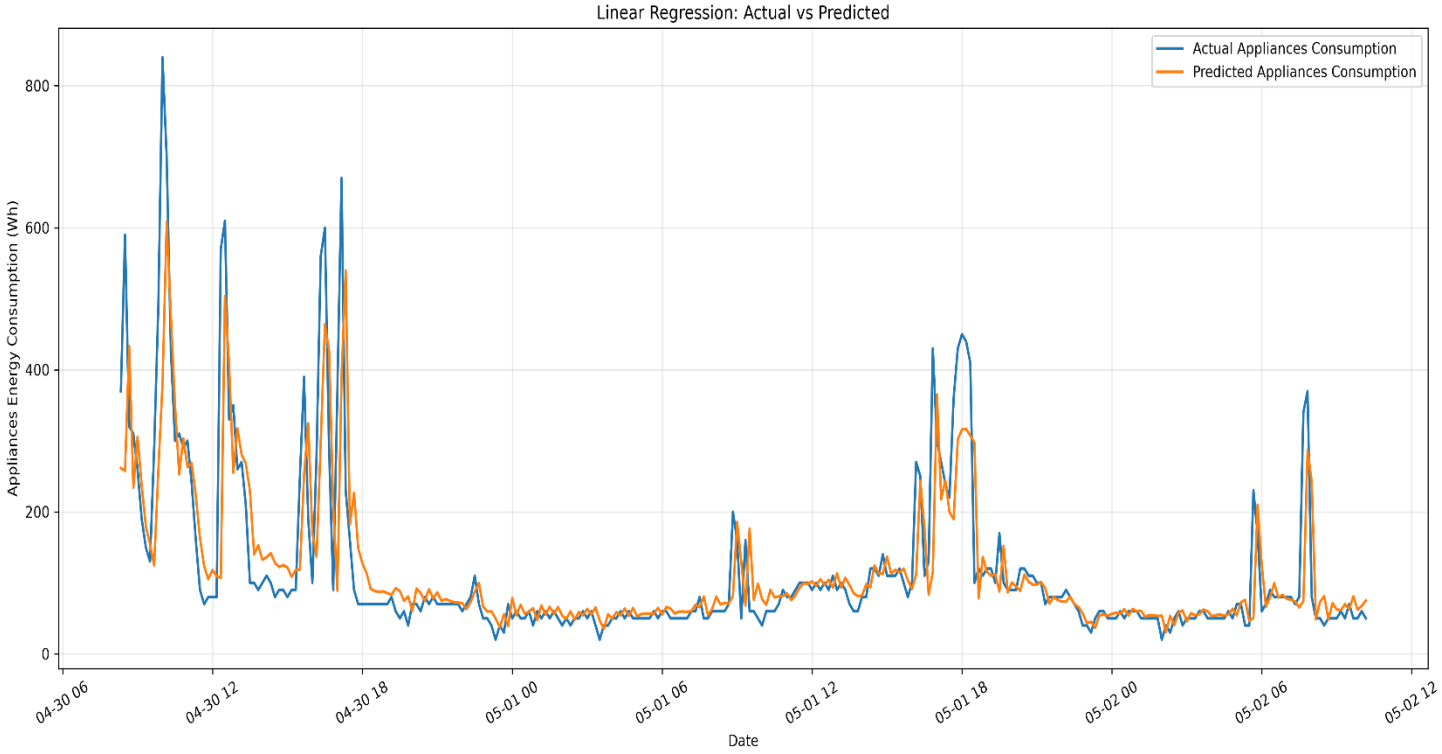
- Input Layer: Receives 3 historical time steps 30 miniute containing the selected input features.

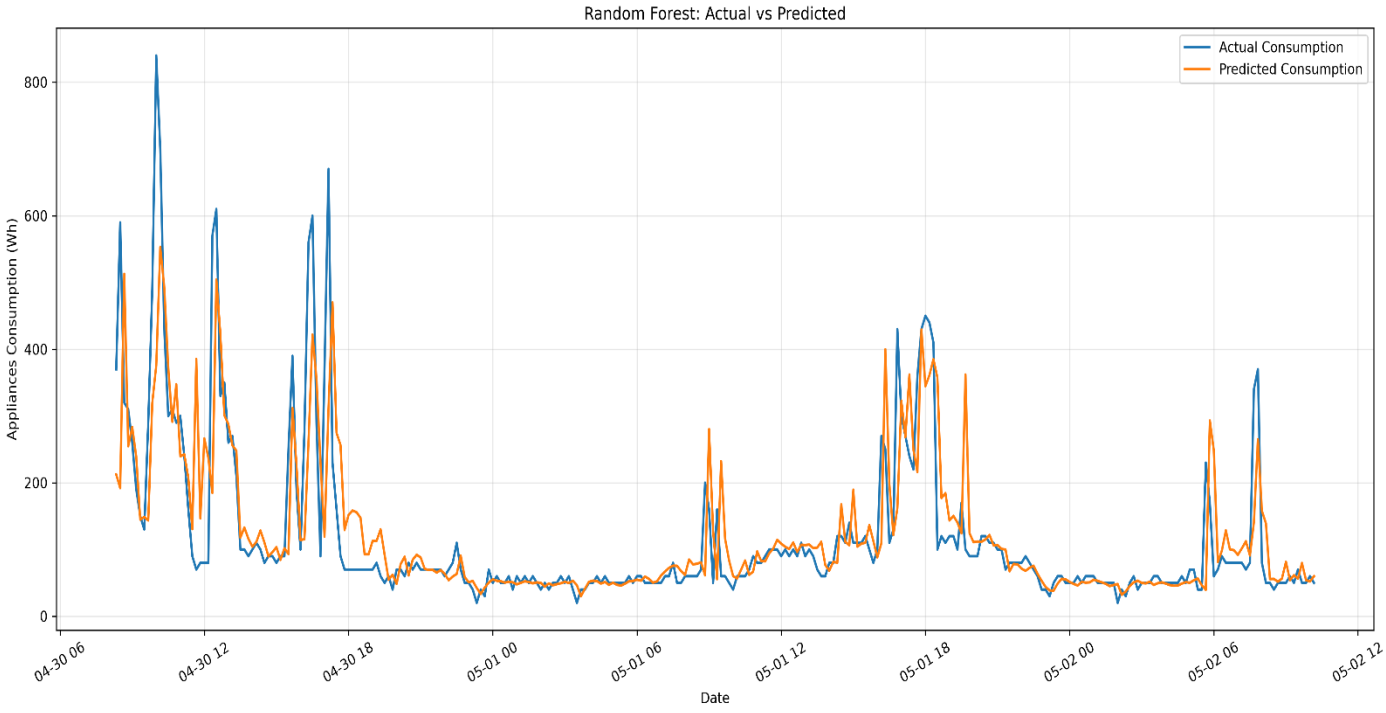
- Conv1D Layers: Extract local patterns and relationships from consecutive time steps using 64 and 32 filters.
- LSTM Layers: Learn temporal dependencies and sequential patterns in the extracted features using 64 and 32 neurons.
- Dropout (0.2): Reduces overfitting by randomly disabling 20% of neurons during training.
- Dense Layer (16 neurons): Learns complex relationships from the extracted temporal features.
- Output Layer (1 neuron): Produces the predicted appliance energy consumption for the next 10-minute interval.

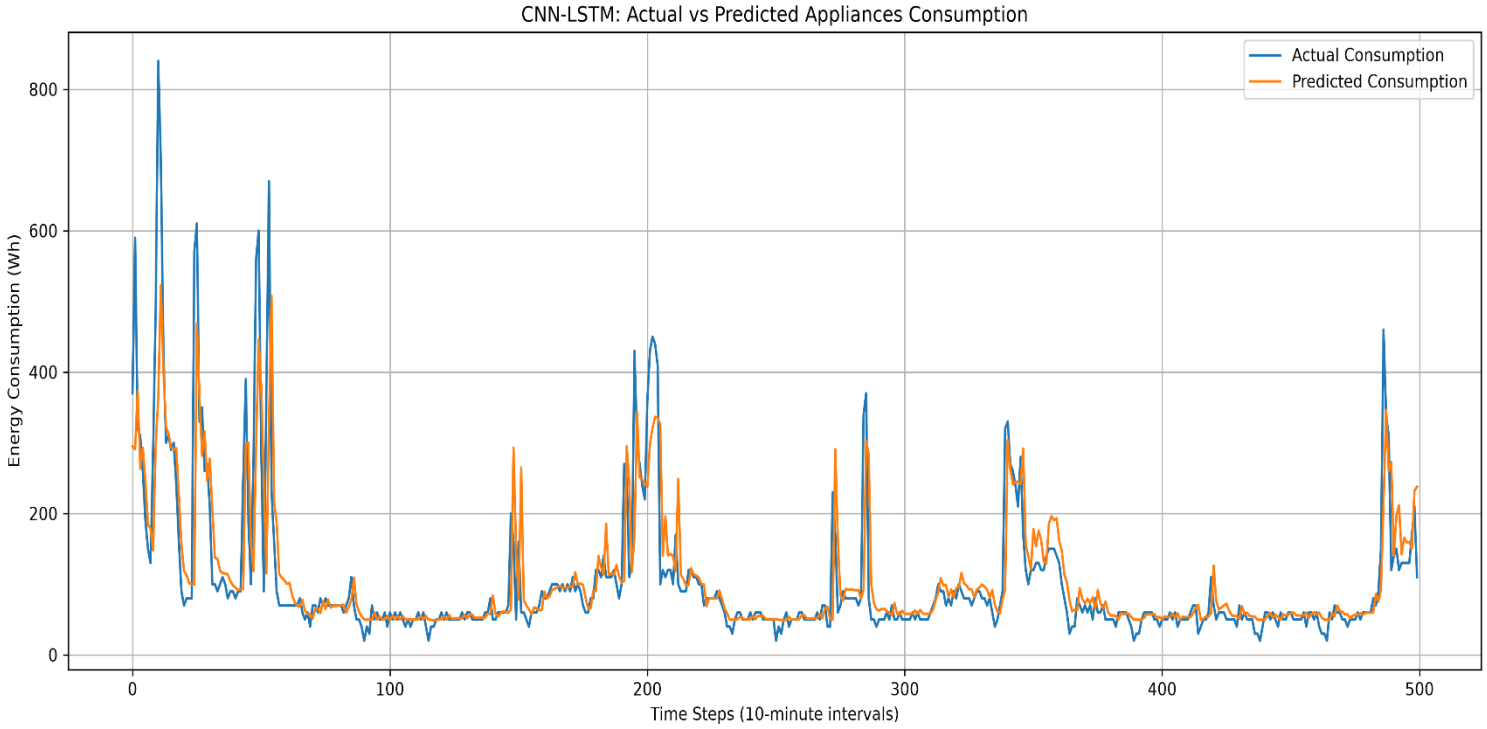
Activation functions

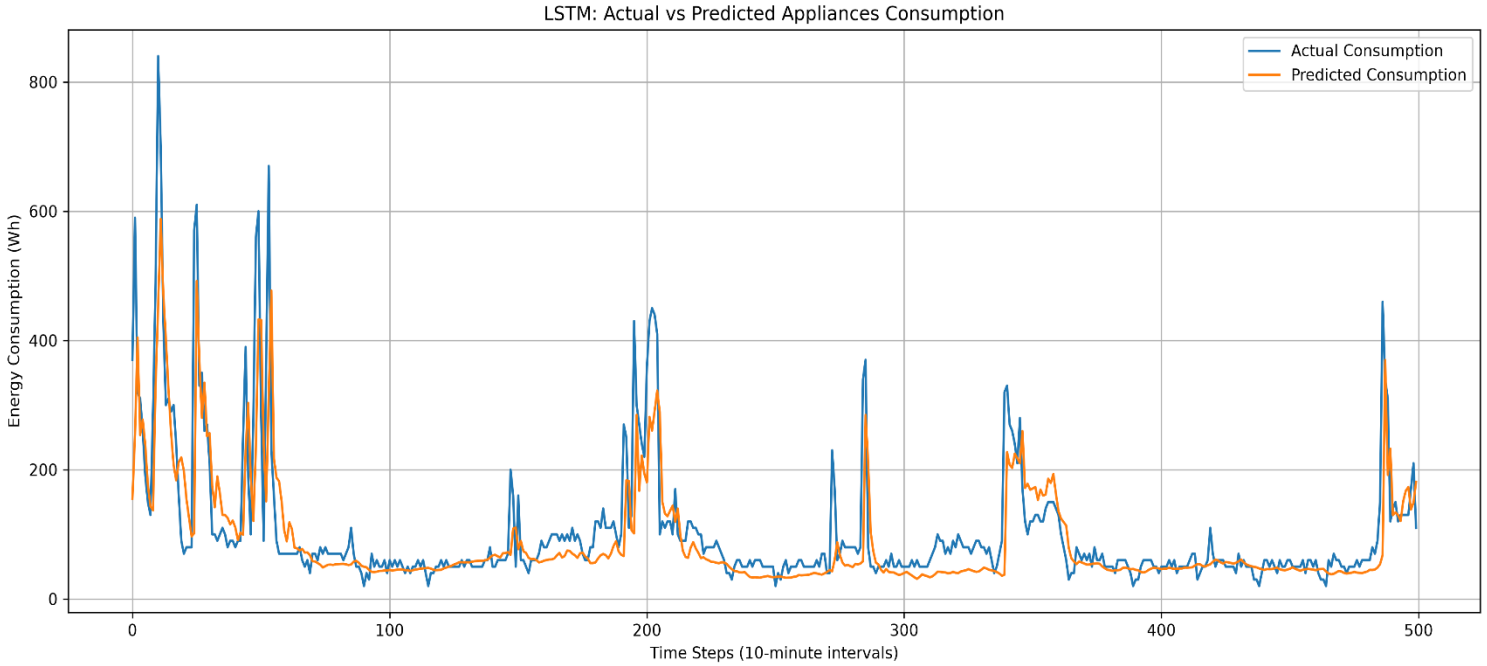
- ReLU: Used in both Conv1D layers and the Dense layer to introduce non-linearity and extract complex patterns.
- Tanh: Used in both LSTM layers to regulate information flow through recurrent units, producing values between −1 and 1.
- Linear: Used in the output layer because the model predicts a continuous numerical value.

Results


Linear Regression: Actual vs Predicted
Actual Appliances Consumption
Predicted Appliances Consumption
Appliances Energy Consumption (Wh)
0
200
400
600
800
04-30 06
04-30 12
04-30 18
05-01 00
05-01 06
05-01 12
05-01 18
05-02 00
05-02 06
05-02 12
Date


Random Forest: Actual vs Predicted
Actual Consumption
Predicted Consumption
Appliances Consumption (Wh)
0
200
400
600
800
04-30 06
04-30 12
04-30 18
05-01 00
05-01 06
05-01 12
05-01 18
05-02 00
05-02 06
05-02 12
Date


CNN-LSTM: Actual vs Predicted Appliances Consumption
Actual Consumption
Predicted Consumption
Energy Consumption (Wh)
Time Steps (10-minute intervals)
0
200
400
600
800
0
100
200
300
400
500


LSTM: Actual vs Predicted Appliances Consumption
Actual Consumption
Predicted Consumption
Energy Consumption (Wh)
Time Steps (10-minute intervals)
0
200
400
600
800
0
100
200
300
400
500

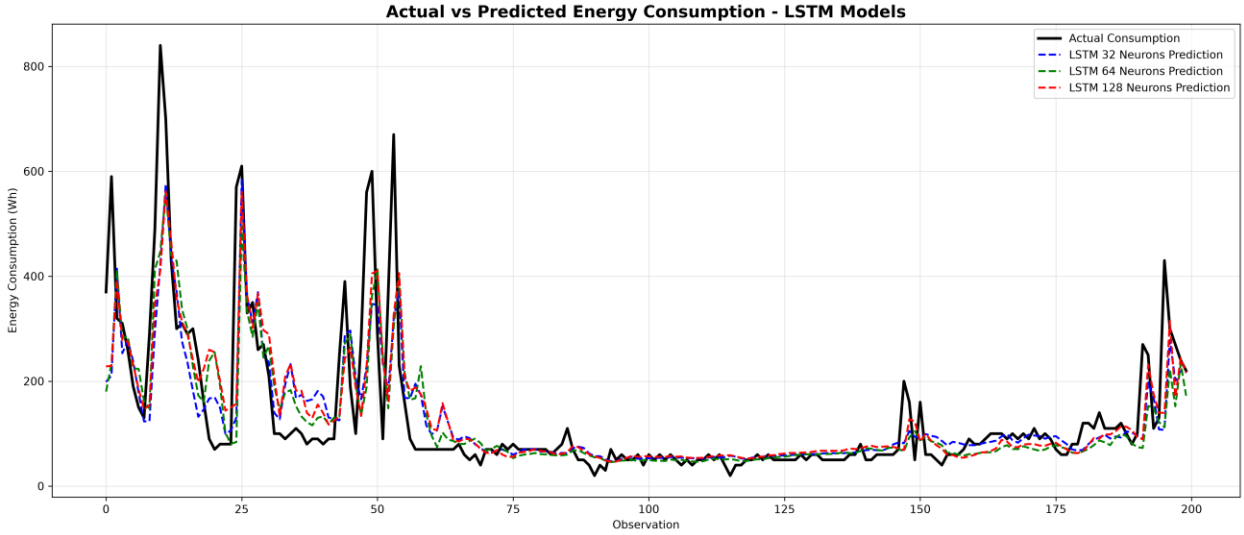
Evaluation Metrics Comparison Against Baseline Models (Linear Regression and Random Forest)

Model	MAE (Wh)	MSE	RMSE (Wh)	R^2	R^2 (%)
Random Forest	42.6029	6023.5359	77.6114	0.2732	27.32%
Linear Regression	27.2182	3599.4786	59.9957	0.5657	56.57%
CNN-LSTM	26.8130	3542.3765	59.5179	0.5726	57.26%
LSTM	28.7445	4088.5903	63.9421	0.5067	50.67%

6.Model Optimization

Model optimization was performed by experimenting with different hyperparameter configurations. The learning rate and dropout rate were adjusted and evaluated to identify their optimal values. Similarly, different batch sizes, numbers of epochs, numbers of layers, and neurons per layer were tested.

The best-performing hyperparameter values were selected based on the evaluation results. The models were then trained using these optimized configurations, and the resulting performance improvements are presented in the diagrams below.



Similarly, all hyperparameters were varied, and the optimal values were selected using Bayesian Optimization. Therefore, the finalized optimized hyperparameter values for each model are as follows:

Parameter	LSTM Model	CNN-LSTM Model
Learning Rate	0.001 (min: 0.0001)	0.001 (min: 0.0001)
Batch Size	64	64
Number of Epochs	50	50
Number of Layers	2 LSTM layers	2 Conv1D + 2 LSTM layers
Neurons / Filters per Layer	128, 128	Conv1D: 64, 32; LSTM: 64, 32

7.Challenges

One of the main challenges during data preprocessing was determining whether the identified outliers represented genuine anomalies or valid data observations. It was difficult to make this distinction based solely on statistical methods. Therefore, understanding the context and characteristics of the dataset was essential. To address this challenge, relevant research papers that used the same dataset were reviewed, and their approaches to handling outliers were considered. Based on the findings from the literature, reasonable decisions were made to distinguish between potential anomalies and genuine observations.

Another challenge was the presence of unusually low values in the RH_6 (Relative Humidity) feature, with some readings as low as 1%. Such values are extremely rare under normal environmental conditions, raising concerns about whether they represented sensor errors or genuine measurements. To investigate this, several research papers that used the same dataset were reviewed. Since most of these studies retained the RH_6 feature during preprocessing, it was also retained in this study. This decision was made based on the existing research, while acknowledging that these unusual readings could potentially represent genuine environmental conditions or sensor-related variations.[3]

8.CONCLUSION

The overall performance evaluation demonstrated that deep learning models can effectively capture patterns and temporal dependencies in household energy consumption data. Both LSTM and CNN-LSTM models were developed and evaluated alongside the baseline models, Linear Regression and Random Forest. Hyperparameter optimization was performed using Bayesian Optimization to identify suitable configurations, including learning rate, batch size, number of epochs, number of layers, and neurons per layer.

The experimental results showed that the CNN-LSTM model achieved the best overall performance, with an MAE of 26.81 Wh, RMSE of 59.52 Wh, and R^2 score of 57.26%. It slightly outperformed Linear Regression and the standard LSTM model, demonstrating the potential benefits of combining CNN and LSTM layers for household energy consumption prediction.

Despite these promising results, several limitations provide opportunities for future improvements. One major limitation of this study is that the analysis was conducted using data collected from a single household. Future research could incorporate data from multiple households to identify more generalizable energy consumption patterns and investigate how factors such as the number and age of occupants, pet ownership, and building characteristics influence appliance energy consumption.

Furthermore, the dataset covers a limited period, which may not fully capture seasonal variations in energy consumption. Collecting data over longer periods would help identify changes in appliance usage patterns across different seasons. The accuracy of predictions could also be improved by using weather measurements collected closer to the target household. In addition, future studies could investigate the optimal placement of indoor wireless sensors, incorporate additional sensors, and use more accurate measurement devices to obtain better-quality data.

Overall, this study provides a foundation for household energy consumption prediction using machine learning and deep learning techniques. Addressing these limitations through larger and more diverse datasets, improved sensor configurations, and extended data collection periods could further enhance prediction accuracy and support more effective household energy management.

9.Reference

[1] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

[2] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” Proceedings of the IEEE, vol. 86, no. 11, pp. 2278–2324, 1998. doi: 10.1109/5.726791.

[3] L. M. Candanedo, V. Feldheim, and D. Deramaix, “Data driven prediction models of energy use of appliances in a low-energy house,” Energy and Buildings, vol. 140, pp. 81–97, 2017. doi: 10.1016/j.enbuild.2017.01.083